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TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

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First 10 records:

	Age	BloodPressure	Cholesterol	Glucose	BMI	FamilyHistory	Diabetes
0	58	105	199	183 22.5	0	1	
1	71	178	214	184 22.5	0	1	
2	48	129	237	83 19.0	1	1	
3	34	104	263	78 37.9	0	0	
4	62	103	127	109 35.9	1	0	
5	27	92	251	94 30.8	0	0	
6	80	139	223	172 40.8	1	1	
7	40	86	251	91 39.8	0	0	
8	58	136	144	73 33.8	0	0	
9	77	115	215	98 41.4	0	0	

Dataset shape:

(500, 7)

Class distribution:

Diabetes

0 320

1 180

Name: count, dtype: int64

Class distribution percentage:

Diabetes

0 64.0

1 36.0

Name: proportion, dtype: float64

Missing values:

Age 0
BloodPressure 0
Cholesterol 0
Glucose 0
BMI 0
FamilyHistory 0
Diabetes 0
dtype: int64

Target variable: Diabetes

0 = Non-Diabetic

1 = Diabetic

Features:

['Age', 'BloodPressure', 'Cholesterol', 'Glucose', 'BMI', 'FamilyHistory']

Training samples: 350

Testing samples: 150

Class distribution BEFORE SMOTE:

Diabetes

0 224

1 126

Name: count, dtype: int64

SMOTE applied successfully.

Class distribution AFTER balancing:

Diabetes

1 224

0 224

Name: count, dtype: int64

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TASK 2: DECISION TREE FOR DIAGNOSIS

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DECISION TREE

Accuracy : 0.5267

Precision : 0.3191

Recall : 0.2778

F1-Score : 0.297

AUC-ROC : 0.4249

Confusion Matrix:

[[64 32]

[39 15]]

True Negatives : 64

False Positives: 32

False Negatives: 39

True Positives : 15

Classification Report:

	precision	recall	f1-score	support
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Non-Diabetic	0.62	0.67	0.64	96
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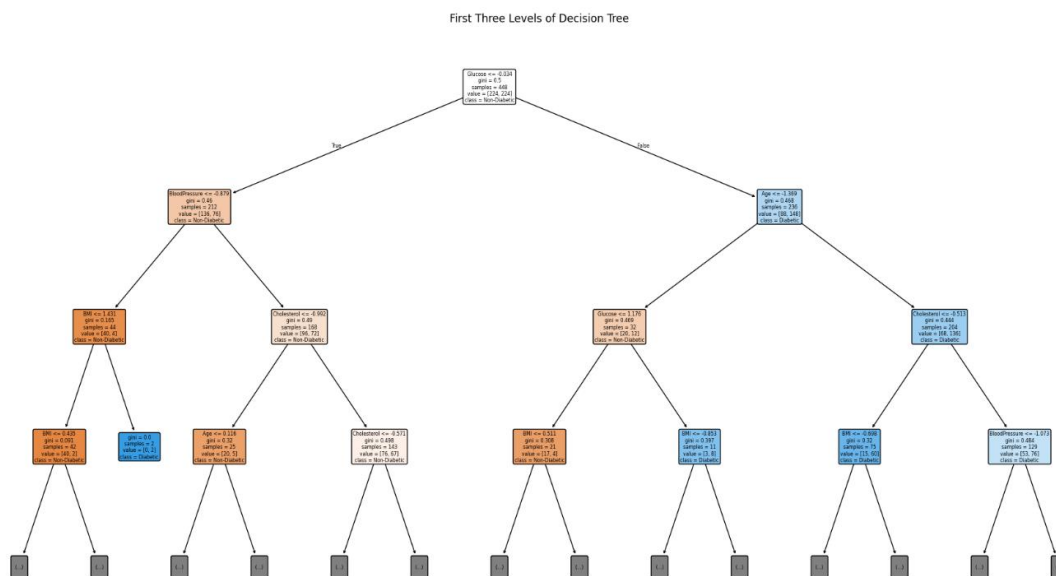
Diabetic	0.32	0.28	0.30	54
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accuracy		0.53	150	
macro avg	0.47	0.47	0.47	150
weighted avg	0.51	0.53	0.52	150

Decision Tree structure:

Maximum depth: 5

Number of leaves: 25



Decision Tree Feature Importance:

Feature Importance

3	Glucose	0.355036
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4 BMI 0.243825

1 BloodPressure 0.174929

2	Cholesterol	0.155983
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0 Age 0.070227

5 FamilyHistory 0.000000

Top 3 Decision Tree Predictors:

Feature Importance

3 Glucose 0.355036

4 BMI 0.243825

1 BloodPressure 0.174929

Decision Tree Root Feature: Glucose

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TASK 2(c): POST-PRUNING

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POST-PRUNED DECISION TREE

Accuracy : 0.52

Precision : 0.3125

Recall : 0.2778

F1-Score : 0.2941

AUC-ROC : 0.5017

Confusion Matrix:

[[63 33]

[39 15]]

True Negatives : 63

False Positives: 33

False Negatives: 39

True Positives : 15

Classification Report:

	precision	recall	f1-score	support
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Non-Diabetic	0.62	0.66	0.64	96
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Diabetic	0.31	0.28	0.29	54
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accuracy		0.52	150
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macro avg	0.47	0.47	0.47	150
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weighted avg	0.51	0.52	0.51	150
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Selected ccp_alpha: 0.004120879120879119

Pruned Tree Depth: 13

Pruned Tree Leaves: 47

Pre-Pruning vs Post-Pruning

	Metric	Pre-Pruned Tree	Post-Pruned Tree
0	Accuracy	0.5267	0.5200
1	Precision	0.3191	0.3125
2	Recall	0.2778	0.2778
3	F1-Score	0.2970	0.2941
4	AUC-ROC	0.4249	0.5017
5	False Negatives	39.0000	39.0000

TASK 3: STATISTICAL LEARNING - LOGISTIC REGRESSION

LOGISTIC REGRESSION

Accuracy : 0.56
Precision : 0.3929
Recall : 0.4074
F1-Score : 0.4
AUC-ROC : 0.5766

Confusion Matrix:

[[62 34]
[32 22]]

True Negatives : 62
False Positives: 34
False Negatives: 32
True Positives : 22

Classification Report:

	precision	recall	f1-score	support
Non-Diabetic	0.66	0.65	0.65	96
Diabetic	0.39	0.41	0.40	54
accuracy		0.56		150
macro avg	0.53	0.53	0.53	150
weighted avg	0.56	0.56	0.56	150

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MODEL PERFORMANCE COMPARISON

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	Metric	Decision Tree	Logistic Regression
0	Accuracy	0.5267	0.5600

1	Precision	0.3191	0.3929
2	Recall	0.2778	0.4074
3	F1-Score	0.2970	0.4000
4	AUC-ROC	0.4249	0.5766
5	False Negatives	39.0000	32.0000

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FEATURE IMPORTANCE ANALYSIS

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Logistic Regression Feature Importance:

	Feature	Coefficient	Absolute_Importance
3	Glucose	0.503392	0.503392
1	BloodPressure	0.320665	0.320665
0	Age	0.264101	0.264101
4	BMI	0.101067	0.101067
2	Cholesterol	-0.065353	0.065353
5	FamilyHistory	-0.026159	0.026159

Top 3 Logistic Regression Predictors:

	Feature	Coefficient
3	Glucose	0.503392
1	BloodPressure	0.320665
0	Age	0.264101

Top 3 Decision Tree Predictors:

['BMI', 'BloodPressure', 'Glucose']

Top 3 Logistic Regression Predictors:

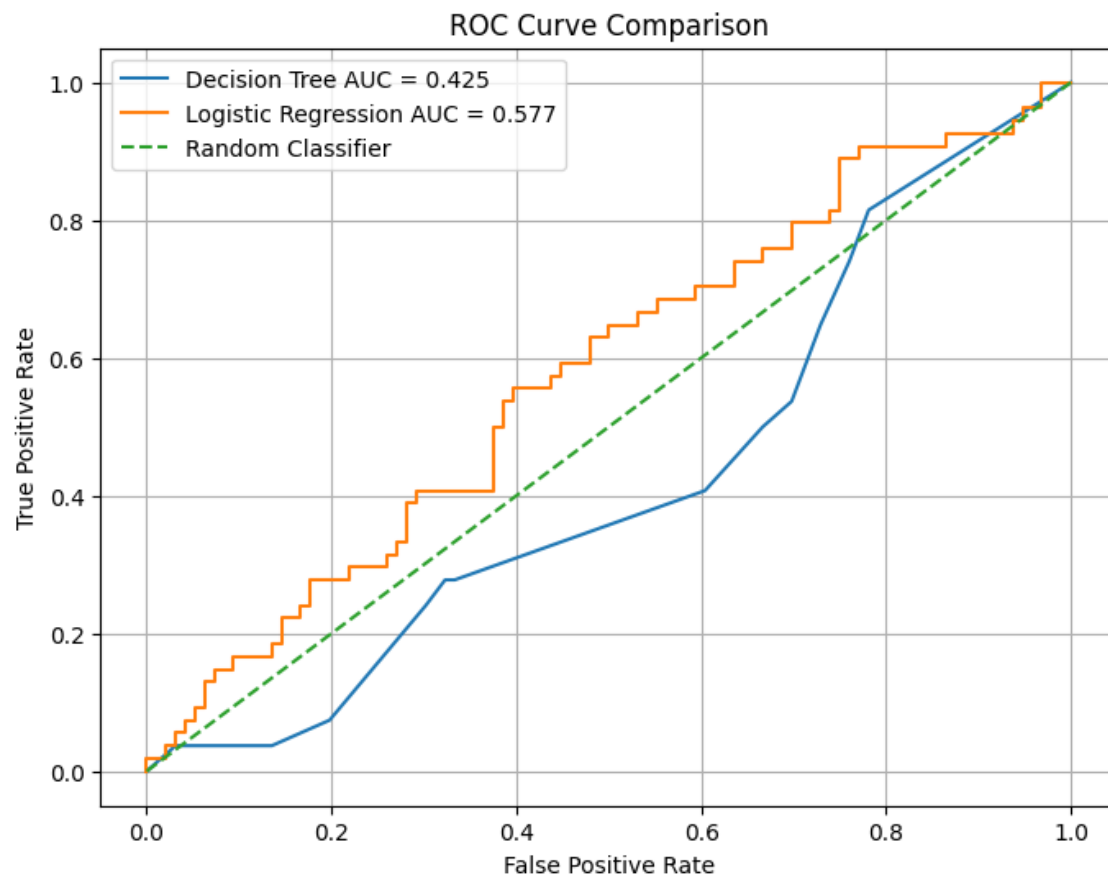
['BloodPressure', 'Glucose', 'Age']

Common Top Predictors:

['BloodPressure', 'Glucose']

Result: Both models show strong agreement among the important predictors.

ROC-AUC COMPARISON



TASK 4: REINFORCEMENT LEARNING FOR TREATMENT

States:

0 = Low Risk

1 = Moderate Risk

2 = High Risk

3 = Controlled

Actions:

0 = Diet

1 = Exercise

2 = Medication

3 = Monitor

Initial Q-Table:

	Diet	Exercise	Medication	Monitor
Low Risk	0.0	0.0	0.0	0.0
Moderate Risk	0.0	0.0	0.0	0.0
High Risk	0.0	0.0	0.0	0.0
Controlled	0.0	0.0	0.0	0.0

Q-Learning Update History:

Episode	State	Action	Reward	Old Q	New Q
0	1 High Risk	Diet	4	0.0000	2.0000
1	1 Moderate Risk	Monitor	2	0.0000	1.0000
2	1 Moderate Risk	Monitor	2	1.0000	1.9500
3	1 Moderate Risk	Medication	3	0.0000	1.5000
4	2 High Risk	Monitor	3	0.0000	2.4000
5	2 High Risk	Monitor	3	2.4000	3.7800
6	2 High Risk	Medication	9	0.0000	4.5000
7	3 Moderate Risk	Monitor	2	1.9500	2.8525
8	3 Moderate Risk	Monitor	2	2.8525	3.7099
9	3 Moderate Risk	Monitor	2	3.7099	4.5244
10	3 Moderate Risk	Diet	6	0.0000	3.0000
11	3 Low Risk	Diet	5	0.0000	2.5000

12	4	Moderate Risk	Monitor	2	4.5244	5.2982
13	4	Moderate Risk	Monitor	2	5.2982	6.0333
14	4	Moderate Risk	Monitor	2	6.0333	6.7316
15	4	Moderate Risk	Monitor	2	6.7316	7.3950
16	4	Moderate Risk	Monitor	2	7.3950	8.0253
17	4	Moderate Risk	Monitor	2	8.0253	8.6240
18	4	Moderate Risk	Monitor	2	8.6240	9.1928
19	4	Moderate Risk	Exercise	7	0.0000	4.6250

Sample Q-Table Updates:

Episode	State	Action	Reward	Old Q	New Q
0	1	High Risk	Diet	4	0.0000 2.0000
1	1	Moderate Risk	Monitor	2	0.0000 1.0000
2	1	Moderate Risk	Monitor	2	1.0000 1.9500
3	1	Moderate Risk	Medication	3	0.0000 1.5000
4	2	High Risk	Monitor	3	0.0000 2.4000
5	2	High Risk	Monitor	3	2.4000 3.7800
6	2	High Risk	Medication	9	0.0000 4.5000
7	3	Moderate Risk	Monitor	2	1.9500 2.8525
8	3	Moderate Risk	Monitor	2	2.8525 3.7099
9	3	Moderate Risk	Monitor	2	3.7099 4.5244

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FINAL Q-TABLE

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	Diet	Exercise	Medication	Monitor
Low Risk	4.8438	0.000	0.7500	3.4688
Moderate Risk	6.6094	4.625	3.6375	14.7296
High Risk	2.0000	0.000	4.5000	3.7800
Controlled	1.5000	0.000	0.0000	0.0000

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FINAL LEARNED POLICY

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Low Risk -> Diet Q-value = 4.8438

Moderate Risk -> Monitor Q-value = 14.7296

High Risk -> Medication Q-value = 4.5000

Controlled -> Diet Q-value = 1.5000

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CONVERGENCE ANALYSIS

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Maximum Q-value change: 8.44125

Convergence threshold: 0.001

Status: More training episodes are required.

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PROJECT SUMMARY

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1. Dataset prepared with healthcare features.
2. Target variable: Diabetes.
3. Dataset split into 70% training and 30% testing.
4. Class imbalance handled using SMOTE/class weighting.
5. Decision Tree trained for diabetes diagnosis.
6. Accuracy, Precision, Recall, F1-score and False Negatives calculated.
7. Decision Tree feature importance calculated.
8. Post-pruning applied using cost-complexity pruning.
9. Logistic Regression trained as a statistical learning model.
10. Decision Tree and Logistic Regression compared using AUC-ROC.
11. Top diabetes predictors identified from both models.
12. Treatment recommendation formulated as an MDP.

13. Q-Learning applied to treatment recommendation.

14. Final Q-table and learned policy generated.

15. Convergence checked using Q-value changes.

IMPORTANT:

The treatment recommendations and Q-Learning environment in this program are educational simulations. They must not be used to make actual medical treatment or medication decisions.

Program execution completed successfully.